**Ohio Northern University**
**ECCS Department**
**<u>ECCS-4441 Ethical Hacking and Penetration Testing</u>** (Spring 2026)
Section 01 – CRN: 34683

***Instructor**: Dr. Ajmal Khan*

| **Instructor Information** | **Class Information** | **Office Hours** |
|---|---|---|
| Office: JLK 234 | Class Time: **Mon: 9:00 – 9:50 am** | Mon: 12 noon – 2 pm |
| *m-khan.2@onu.edu* | **Thu 9:00 – 10:50 am** | Wed: 12 noon – 2 pm |
| | Class Location: **JLK-205** | Thu: 12 noon – 1 pm |
| | | Any other time by appointment |

**Catalog Description** (3 credits)
This course provides a comprehensive introduction to ethical hacking and penetration testing methodologies. Key topics include reconnaissance, scanning, exploitation, post-exploitation, and reporting. Hands-on labs cover real-world penetration testing tools including Kali Linux, Metasploit, Nessus, OSINT, Nmap, Burp Suite, and more. By the end of the course, students will gain practical experience in identifying vulnerabilities, conducting penetration tests, and strengthening system security.
<u>Prerequisite:</u> ECCS 3631 Minimum Grade of D. Must be enrolled in one of the following Classifications: Junior, Senior.

**Course Information**
All course materials, including grades, are available at Canvas: https://onu.instructure.com/courses/14682 (requires login).

**Textbook: (Primary)**
- Robert S. Wilson, *CompTIA PenTest+ Guide to Penetration Testing*, 1st Edition, Cengage, 2024. ISBN: 978-0-357-95065-4.

**Textbooks: (Alternative)**
- Robert S. Wilson, *CompTIA PenTest+ Guide to Penetration Testing*, 2nd Edition, Cengage, 2027. ISBN: 978-8-214-41011-1.
- Mike Chapple, Robert Shimonski, and David Seidl, *CompTIA PenTest+ Study Guide: Exam PT0-003*, 3rd Edition, Sybex, 2025. ISBN: 978-1394285006

**Software Packages:**
- VMWare
- Kali Linux
- Windows Server
- Windows 10
- DVWA
- Metasploitable 2.0
- Metasploitable 3.0
- Windows 7

**Course Fee:**
There is no course fee.

**Course Outcomes:**
After successful completion of this course, the student should be able to:
1. Describe the penetration testing process, cybersecurity concepts, and ethical hacking mindset.
2. Describe the tools used in the testing lab
3. Apply passive reconnaissance techniques
4. Apply active reconnaissance techniques
5. Execute vulnerability scans using different tools and analyze the results
6. Exploiting a vulnerable service using network penetration tools
7. Explain methods and tools used in the exploitation
8. Apply methods and tools used for performing network attacks
9. Explain web application attacks
10. Explain the tools and methods used to compromise WPS, WEP, WPA, and WPA2 wireless security protocols

**Course Contents:**
The following is a tentative list of the contents of this course:
- Module 1: Introduction to Penetration Testing
- Module 2: Penetration Testing Lab Setup
- Module 3: Passive Reconnaissance Techniques
- Module 4: Active Reconnaissance Techniques
- Module 5: Vulnerability Scanning
- Module 6: Network Penetration Testing
- Module 7: Exploitation Methods and Tools
- Module 8: Network Attacks
- Module 9: Exploiting Web-Based Applications
- Module 10: WiFi Hacking

**Grading Policy:**

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|---|---|
| Class-Work, Class Rules, Attendance: | 15% |
| Projects Implementation: | 25% |
| Report Writing: | 20% |
| Quizzes (Four): | 20% |
| Final Exam: | 20% |
| Academic Dishonesty | –10% |
| Phone/Games/Online Chat | –5% |

A: 90-100, B: 80-89, C: 70-79, D: 60-69, and below 60 is F.

The final exam for this course is not currently scheduled on the Registrar's website. A mutually convenient time will be determined in consultation with the students.

**Class Rules:**

Attendance: Attendance will be taken regularly. If you have to miss a class, please let the instructor know at your earliest. If you have any flu-like symptoms, please follow ONU Health and Safety Policy.

Honesty: If you copy any assignment, a 10% penalty will be applied to your overall grade, which will bring your final grade one letter down.

Cell Phones: Cell phones are not permitted during class or lab sessions. Any use of a cell phone will result in a 5% deduction from your overall grade. To preserve the learning environment, all phones must be placed in the Faraday bag at the start of each session.

AI Policy: The use of AI tools is permitted in this course, but students are fully responsible for the accuracy and validity of any AI-assisted work. Any use of AI must be clearly disclosed and referenced. Submissions must reflect the student's own understanding, and errors remain the student's responsibility. The instructor will assess work based on its correctness and quality, but will not spend time identifying errors attributable to AI use.

Canvas: All projects, reports, and quizzes are turned in to Canvas. *Submissions through email will not be accepted at all and will not be responded to.*

Late Policy: There is a late penalty of 20% per day for all assignments. No submission will be accepted by the Canvas system two days after the due date.

Lab Clean up: If instructed, after hands-on work, each student/group is responsible for removing their cables and cleaning up the configurations made to PCs. Properly shut down the Virtual Machines. Properly logout the desktop PC. Failing to do so, a penalty can be applied to your project grades.

**ONU Common Course Policies**

Ohio Northern University is dedicated to providing an equitable educational experience for all enrolled students. Universal course policies applicable to all courses can be found at the following link: https://my.onu.edu/registrars_office/policies.